Analysis of the radiation doses and activity of a ^{137}Cs probe

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In this experiment, we worked with a ^{137}Cs probe and measured its equivalent dose rate for two distances between detector and probe: $\dot{H}_{0cm} = (6.90 \pm 0.46)\,\mu\mathrm{Sv\,h^{-1}}$ and $\dot{H}_{60cm} = (0.460 \pm 0.081)\,\mu\mathrm{Sv\,h^{-1}}$. The radiation during the experiment was harmless concerning the annual limit. Furthermore, we determined the threshold voltage of the used counter tube to be $(512.0 \pm 4.0)\,\mathrm{V}$. We also studied the counter's responsivity which showed a dependence on the distance to the probe. Additionally, we measured the half-life of ^{137m}Ba with $T_{1/2} = (152.3 \pm 1.4)\,\mathrm{s}$, which is consistent with the literature value of 153.12 s. Lastly, we conducted around 1000 measurements of decays with gate times of 1 s and 10 s. The chi-squared tests for a description of the decays with a Poisson or normal distribution could be accepted in all cases.

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1 Introduction

Radioactivity plays an important role in many scopes of application, e.g. energy capture or cancer treatment. Concomitantly, the radiation protection sets clear conditions and threshold values for the population and employees, who are exposed to radiation, concerning the exposure within specific time periods. In this experiment we measure the radiation dose and activity of a ^{137}Cs probe, as well as the half life of a ^{137m}Ba probe. Furthermore, we study the application of a counter tube and its properties.

2 Dosimetry and radiation exposure

At the beginning of this experiment, we installed a dosimeter for longterm readings of the local equivalent dose and noted the measured value every 15 minutes. This ensures a sufficient documentation of the radioactivity in the course room. The threshold value of 5 mSv during one year must not be exceeded [1].
The used substance was a γ-radiation emitting ^{137}Cs probe with a *half-life* of $T_{1/2} = 30.2\,\mathrm{a}$ [1]. The photon energy of the emitted radiation is $E = 661\,\mathrm{keV}$. Furthermore, the probe is installed into a lead cube and closed with an iron stopper. According to the information at the working station, the probe *S 4239* was produced on the 21st of August 1978 with an *activity* of $A_0 = 3.70\,\mathrm{MBq}$. The activity follows an exponential decay and describes the number of decays per second:

$$A(t) = A_0 e^{-\lambda t} \tag{1}$$

Additionally, the relation between *half-life* $T_{1/2}$ and the *decay constant* λ is given by:

$$\lambda = \frac{\ln(2)}{T_{1/2}} \tag{2}$$

With the elapsed time since the production of 15927 days, eq. 1 and 2 yield:

$$\lambda = 0.023\,\mathrm{a}^{-1}$$
$$A(15927d) = 1.36\,\mathrm{MBq}$$

For the first experiment, we need further definitions [3], like the *absorbed dose* D, which describes the transferred energy E_D into a material with mass m, and is defined by:

$$D = \frac{dE_D}{dm} \tag{3}$$

In terms of radiation protection, another related quantity is the *equivalent dose* H. It represents the health effects of radiation on the human body. The *weighting factor* q distinguishes between different kinds of ionizing radiation and ranges from 1 (β-

and γ-radiation) to 20 (α-radiation). They are linked to the absorbed dose by the equation:

$$H = qD \tag{4}$$

We used the dosimeter *RADEX RD 1503* to measure the radiation in the course room during the experimental procedure. Therefor, we noted the *equivalent dose rate* $\dot{H}$ every 15 minutes. Under the assumption that we can expect β- and γ-radiation only, we set $q = 1$. Furthermore, we assume a constant radiation, so that the equivalent dose for a time period t can be computed with:

$$H(t) = \dot{H}t \tag{5}$$

where $\dot{H}$ is the mean value of the measured equivalent dose rates. This yields the following result for the background radiation over 4 hours:

$$H_{bg,4h} = (0.414 \pm 0.022)\,\mu\text{Sv}$$

The error was calculated as standard error of the series of measurements during 4 h. Transferring this result to one year, leads to an expected equivalent dose of

$$H_{bg,1a} = (907 \pm 48)\,\mu\text{Sv}$$

which corresponds to 18 % of the legally permitted 5 mSv as annual limit value.
To measure the equivalent dose rate of the ^{137}Cs probe, we used the *Neva* dosimeter and two different distances between detector and probe. For both distances, we executed a series of measurements and computed the mean values and standard errors as the device error was not representative for the observed fluctuations.

$$\dot{H}_{0cm} = (6.90 \pm 0.46)\,\mu\text{Sv}\,\text{h}^{-1}$$
$$\dot{H}_{60cm} = (0.460 \pm 0.081)\,\mu\text{Sv}\,\text{h}^{-1}$$

We calculated the equivalent dose for 4 h as well:

$$H_{0cm,4h} = (27.6 \pm 1.8)\,\mu\text{Sv}$$
$$H_{60cm,4h} = (1.84 \pm 0.32)\,\mu\text{Sv}$$

The permitted annual limit value will be achieved after $(1.09 \pm 0.16) \cdot 10^4$ h in close proximity (Assuming H_{60cm}). During the experiment the probe emitted a radiation of (0.0368±0.0065) % of this value. Furthermore, we computed the the probe's radiation adjusted for the background radioactivity, which yields:

$$\dot{H}'_{60cm} = (0.356 \pm 0.081)\,\mu\text{Sv}\,\text{h}^{-1}$$

3 Working with a Geiger-Müller counter

A Geiger-Müller counter is an instrument used for detecting ionizing radiation. In this experiment we study three properties of a counter tube.

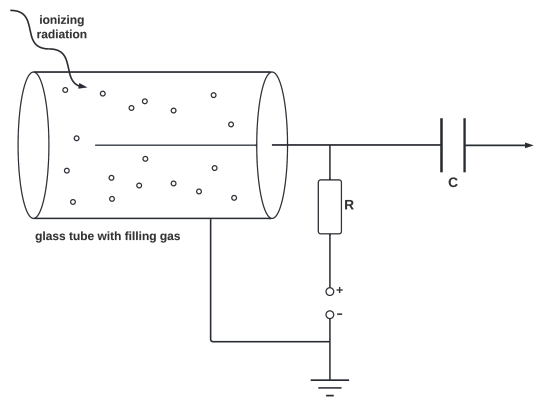


Fig. 1: Schematic structure of a Geiger-Müller-tube. Adapted from [1]

3.1 Threshold voltage

Fig.1 shows the setup of a counter tube. For the measurement, the counter was positioned above the lead cube's mouth. We applied different potentials, used a gate time of 60 s and counted the measured impulses. According to the theory, we expect a larger number of detected decays for a higher voltage until saturation is achieved. γ-radiation produces free electrons inside the filling gas of the tube. These electrons are accelerated by the applied voltage between barrel (anode) and a wire in the tube's center (cathode), which leads to collision ionization of further electrons, called electron avalanche. For this process a minimal threshold voltage is necessary and shall be determined in this part.

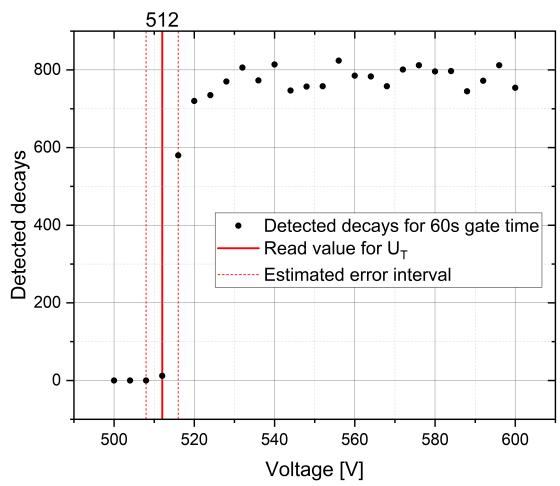


Fig. 2: The plot shows the measured characteristic curve of the used Leybold counter. We determined the threshold voltage to be (512.0 ± 4.0) V (red line). After this value, one recognizes a plateau value in the decays.

The characteristic curve (fig.2) of the *Leybold 57548* counter shows, that the threshold voltage represents:

$$U_0 = (512.0 \pm 4.0)\,\mathrm{V}$$

Here, the error is given by the device's adjustability, as the voltage could only be manipulated in 4.0 V steps.

3.2 Background radiation

To determine the radiation of a specific probe we need to know the background radiation caused by cosmic radiation or other naturally occurring radioactivity. For this experiment we removed all radioactive probes and counted 234 decays during 5 min.
Next, we repeated the measurement and installed the detector tube inside a lead shielding. In the same time period, only 89 impulses were detected, which corresponds to a decrease of 62 %. This implies a distinct shielding effect of the lead. However, the radiation inside the shield is not minimized perfectly. This could be explained by the shield not fully enclosing the counter and the natural occurrence of the radioactive ^{210}Pb in lead.

3.3 Response characteristic of a Geiger-Müller counter

The response characteristic of a counter tube is the probability that a particle, striking the tube is also detected by the counter, as not every photon interacts with the filling gas, producing an electron.
For the measurement, we took the ^{137}Cs probe out of the lead cube and positioned the Geiger-Müller counter at diverse distances d. We set a gate time of 10 s and counted the measured photons. The detector masks a solid angle, described by:

$$\Omega = \frac{a_{SC}}{d^2} \tag{6}$$

According to [1], the counter has an effective surface of $a_{SD} = 11\,\mathrm{cm}^2$. This leads for the furthest $d = (85.00 \pm 0.50)\,\mathrm{cm}$ and nearest distance $d = (5.00 \pm 0.50)\,\mathrm{cm}$ to:

$$\Omega_{85} = (152.2 \pm 1.8) \cdot 10^{-5}\,\mathrm{sr}$$
$$\Omega_5 = (0.440 \pm 0.088)\,\mathrm{sr}$$

The error was calculated with Gaussian error propagation of the distance uncertainty. To compute the responsivity, we used the following formula:

$$R = \frac{4\pi N}{\Omega A}, \tag{7}$$

in which the count of detected particles per second is represented by N. We determined the responsivity to be:

$$R_{85} = (1.214 \pm 0.014)\,\%$$
$$R_5 = (0.72 \pm 0.14)\,\%$$

The obtained, small value of R implies that the counter tube misses many decays because of the probability of an interaction between photon and gas molecules seems to be small. Furthermore, we recognized a trend in the measured data, that the responsivity decreases while reducing the distance. Hence, the assumption of a homogeneous, spherical radiation is probably too simple which explains the apparent decrease of R. For homogeneous radiation, this quantity should be constant as it is a device property.

4 Half-life of ^{137m}Ba

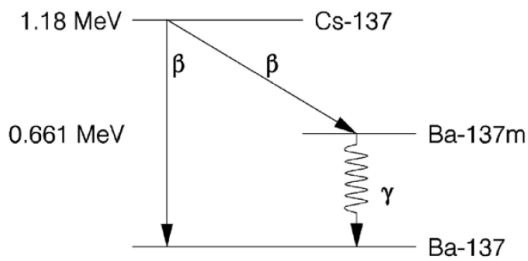


Fig. 3: There are two possible decay options for ^{137}Cs: Either it decays directly into the ground state ^{137}Ba (left hand side , with 0.61 MeV or it decays firstly into ^{137m}Ba (right hand side), an excited state which then leads to the emission of a photon. Carried from [1]

Fig.3 outlines two possible β-decays of ^{137}Cs. Either it decays directly into the ground state ^{137}Ba or it firstly decays into ^{137m}Ba, an excited state which leads to the emission of a photon. In this experiment, we are interested in the half-life of ^{137m}Ba. The detector was installed in front of the probe and the gate time was set to 10 s. Fig.4 visualize the measured data. In general the activity of a radioactive probe can be described by:

$$A(t) = A_0 e^{-\frac{t}{\tau}} + A_{bg} \tag{8}$$

with

$$\tau = \frac{1}{\lambda} \tag{9}$$

,where A_{bg} represents the background activity and λ the decay constant. We fitted the measurement with an exponential function, this yields for $T_{1/2}$:

$$T_{1/2} = (152.3 \pm 1.4)\,\mathrm{s}$$

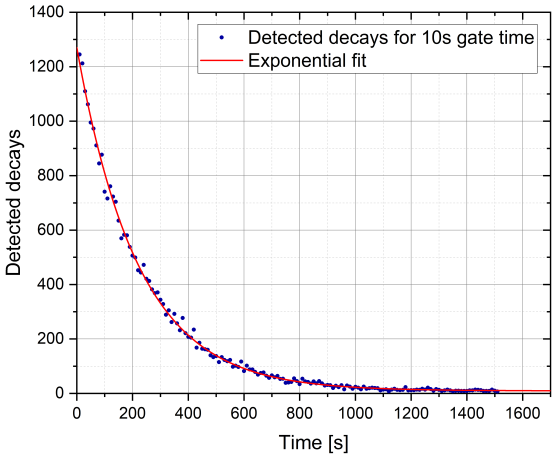


Fig. 4: This plot visualizes the exponential decay. With a fit in the form of eq.8, we computed the half-life of ^{137m}Ba. The determined value of (152.3 ± 1.4) s is consistent with the literature value [2]

The error was computed with Gaussian propagation of uncertainty of the fit parameter's error. This value is consistent with the literature value of 153.12 s [2].

5 Counting statistics of a radioactive decay

In the experiment, it is important to note the stochastic character of the examined decays. Measured data will scatter around its prediction. To quantify the randomness of a radioactive decay, we study the description of a radioactive process with a Poisson and normal distribution.

5.1 Poisson and normal distribution

Actually, the probability of a decay during a fixed time follows a binomial distribution with the parameters N (number of atoms) and the probability of one single decay p. However, in the limit value consideration of $N \to \infty$ and $p \to 0$ with a constant mean $\mu = pN$ we can use the Poisson distribution as an approximation. The Poisson distribution

$$P(x) = \frac{\mu^x e^{-\mu}}{x!}, \tag{10}$$

with x as the number of a specific event's occurrences, and μ the event's expected value, describes the discreet distribution of rare events independently of the time since the last event.
For an expected value $\mu = pN > 9$, we can approximate the Poisson distribution with a normal distribution:

$$G(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}} \tag{11}$$

with

$$\sigma = \sqrt{\mu} \tag{12}$$

which is the Poisson error.

5.2 Statistical hypothesis testing

In this part, we measured the radiation of an uranium oxide plate with a Geiger-Müller counter. For the first series of measurements, we set the gate time to 1 s and recorded 1002 decays. Fig.5 depicts the histograms of the measured data and the expected values for a Poisson distribution. The mean value and its Poisson error represent:

$$\mu_1 = (0.79 \pm 0.89)$$

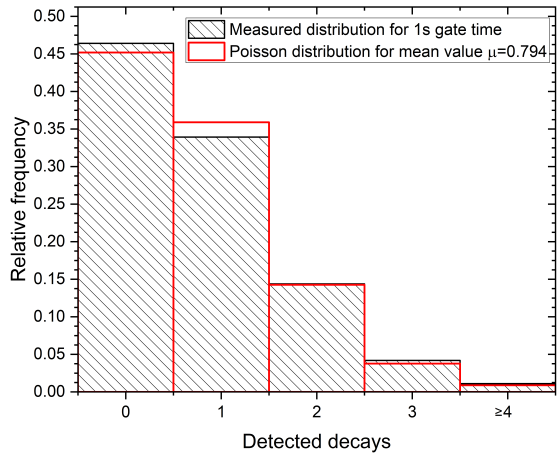
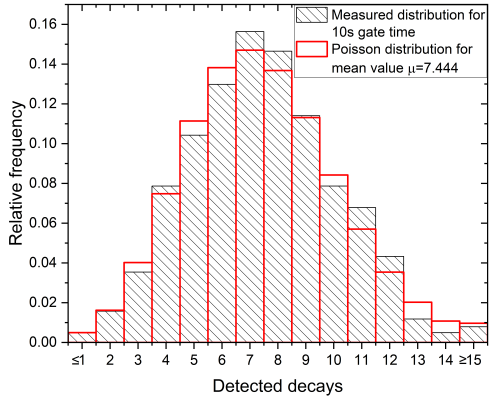


Fig. 5: The gray histogram shows the measured decay distribution with a gate time of 1 s. We plotted the theoretical course (red) of a Poisson distribution with the mean value of the measurements, for comparison. Altogether, we found a distinct congruence, visually and mathematically, with $\chi^2_{P,1} = 2.38$.

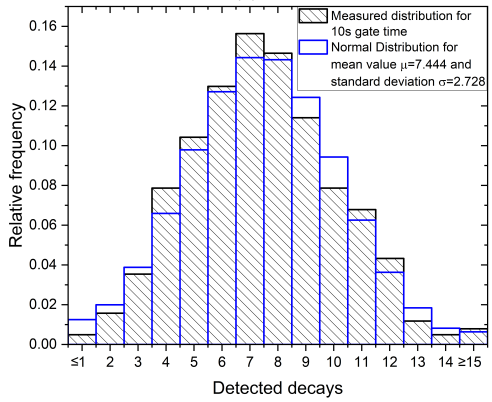
We had to concentrate some events into one class to satisfy the condition of a chi-squared test that the theoretical value for each class has to be greater than five. Consequently, all events with more than three decays are described by class 4. For the acquired data, we calculated the chi-square to:

$$\chi^2_{P,1} = 2.38$$

At a significance level of 5 % for three degrees of freedom, the threshold is $\chi^2_{P,5\%} = 7.81$ [4]. Since $\chi^2_{P,1}$ is smaller than the threshold, we accept the hypothesis of a Poisson distribution with a significance level of 5 %.
Next, we tested 1017 datapoints with a gate time of 10 s for a Poisson and normal distribution. The results are plotted in fig.6. We determined the mean



(a)



(b)

Fig. 6: Here, one can see the 10 s gate time measurement (gray), in 6a compared with a Poisson , and in 6b compared with a normal distribution. We calculated $\chi^2_{P,10} = 14.52$ and $\chi^2_{N,10} = 19.40$ which lead to an acceptance of the hypothesis of a Poisson or Gaussian distribution.

value and its error to be:

$$\mu_{10} = (7.4 \pm 2.7)$$

Subsequently, we calculated the following chi-squares:

$$\chi^2_{P,10} = 14.52$$
$$\chi^2_{N,10} = 19.40$$

With 13 degrees of freedom for the Poisson test and with 12 degrees of freedom for Gaussian test, we looked up the threshold values of χ and found $\chi^2_{P,5\%} = 22.36$, as well as $\chi^2_{N,5\%} = 21.03$ [4]. Finally, we compared these values with the ones we calculated before. Both thresholds lie above the calculated values. Therefore we can accept the hypothesis of a Poisson and Gaussian distribution as well.
Fig.7 is a probability plot of the 10 s-measurement. Additionally, we fitted the cumulated probabilities with a line, at which a perfect accordance would imply a Gaussian distribution of the data. The pearson correlation coefficient for the fit represents 0.9974.

Since a value of 1 is desirable, we suppose that our data follow a normal distribution. To determine μ, we read the crossing point of the fit line with the 50 % axis. Additionally, we used the definitions of the 1σ-interval, which means that we can expect 68.27 % of the measured data around the expected value. We

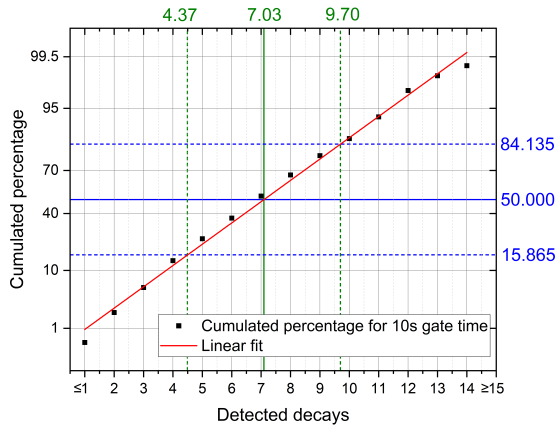


Fig. 7: This probability plot of the 10 s-measurement shows a distinct congruence of the data with a normal distribution. We read the values of μ and σ, using the depicted lines in the graph.

read the crossing points at the ends of this symmetric interval, which gave us the following mean value with error:

$$\mu_{10,g} = (7.03 \pm 2.67)$$

We used the mean of the determined values of σ. This expected value is compatible with the earlier computed Poisson error (cf. 12). The deviations represent 5.57 % for μ and 0.58 % for σ.

6 Conclusion

In this experiment, we studied the radiation of a ^{137}Cs probe with regard to the activity, computed to 1.36 MBq and the equivalent dose, determined to be $H_{0cm,4h} = (27.6 \pm 1.8)$ µSv. Additionally, we measured the equivalent dose of the background radiation as well, with $H_{bg,4h} = (0.414 \pm 0.022)$ µSv and calculated that this value undercuts the annual legally permitted threshold of 5 mSv definitely.
Furthermore, we worked with a Geiger-Müller counter. The minimal required voltage, so that the counter detects radiation, represents $U_0 = (512.0 \pm 4.0)$ V. Besides, we studied the background radiation with the counter tube as well. During 5 min we counted 234 decays with no shielding, but only 89 counting impulses when we installed the detector inside a shield, which corresponds to a decrease of 62 %. The last examined property of counter tube was the responsivity R. It describes the probability

that a particle, striking the tube is also detected by the counter, as only some photons interact with the filling gas. For two distances of 85cm and 5cm, we calculated the responsivity to $R_{85} = (1.214 \pm 0.014)\,\%$ and $R_5 = (0.72 \pm 0.14)\,\%$. Our series of measurements indicates a decline of R, while reducing the distance between detector and probe. We suppose that the sample does not radiate homogeneously, spherically.
In the last part of the experiment we counted approximately 1000 decays, firstly with a gate time of 1 s and then 10 s. We examined the descriptiveness of the decays by a Poisson distribution (1 s, 10 s) and a Gaussian distribution (10 s) with a chi-squared test. In the case of 1 s gate time we accepted the assumption of a Poisson distribution. For the measurement with 10 s gate time, both χ laid within the acceptance boundaries, which shows a transition between the Poisson and normal distribution.

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